Transforming Commercial Fleet Operations Through Predictive Intelligence

- Protecting drivers and eliminating stressful roadside breakdowns.
- Moving from rigid calendar rules to smart real-time vehicle monitoring.
- Maximizing fleet availability while protecting client SLAs and margins.

Presented by Aditya Singh Panwar

Understanding the Operational & Human Cost of Breakdowns

- **Driver Safety First:** Roadside mechanical failures put our drivers in hazardous traffic situations and create high job stress.
- **Client Trust Damage:** Delayed shipments erode customer relationships and lead to severe contract penalties.
- **Compounding Costs:** Each incident results in compounding costs due to emergency towing, rush parts, and overtime labor.

Evaluating Options: Why We Chose Predictive AI

OPTION A: STATUS QUO

Reactive Repairs

Fix components only after they break on the highway.

Rejected: Expensive & dangerous

OPTION B: CALENDAR RULES

Rigid Odometer Schedules

Replace parts every 10,000 miles regardless of actual wear.

Rejected: Wastes budget & misses failures

RECOMMENDED SOLUTION

Predictive Intelligence

Listen to real-time engine health signals to fix wear early.

Selected: Lowest cost & highest safety

Listening to Vehicle Health Signals in Real-Time

- **Continuous Monitoring:** Vehicle sensors continuously measure heat, vibration, and pressure - like a continuous pulse check.
- **7-Day Early Warning:** Flags subtle internal strain up to a week before a component fails.
- **Targeted Maintenance:** Shop mechanics receive a clear, priority alert to replace specific worn components during routine stops.



Real-Time Telemetry Processing

Continuous evaluation of engine stress and temperature patterns

From Digital Vital Signs to Shop Action

1. Reading Vital Signs

Sensors track subtle temperature and pressure shifts while driving.

2. Detecting Hidden Wear

Software identifies early warning patterns invisible to human inspection.

3. Simple Mechanic Action

Generates an easy-to-read service ticket with plain recommendations.

Articulating Model Results for Business Leadership

- **8 in 10 Failures Caught Early:** Across holdout test evaluations representing thousands of operational machine cycles, the model accurately flagged 82% of impending breakdowns with a 0.2% false alarm rate.
- **Minimal False Alarms:** False warnings kept within 0.2%, meaning mechanics spend zero time on unnecessary maintenance.
- **7-Day Lead Time:** Gives logistics teams ample time to re-route freight and schedule shop maintenance cleanly.

MODEL ACCURACY & RELIABILITY BENCHMARK

82%

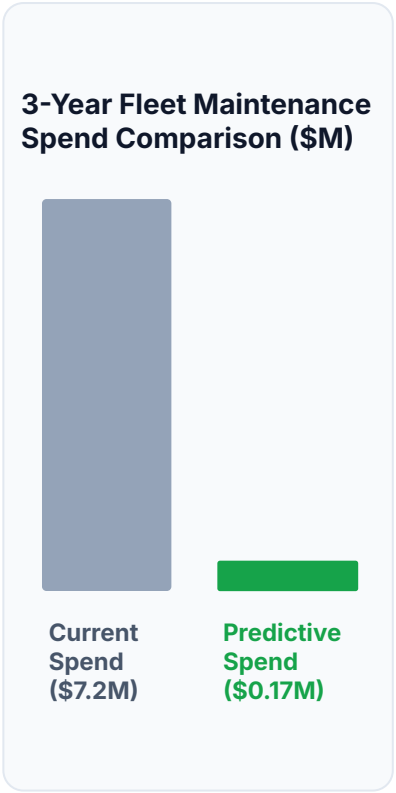
Breakdowns Caught Early

0.2%

False Alarm Rate

Measurable Financial & Operational Return

- **$2.3M Net Annual Savings:** Reduced spend on towing, rush freight shipping, and emergency overtime.
- **28% Capacity Boost:** More active trucks on the road fulfilling client orders instead of sitting in repair bays.
- **Rapid Payback:** Entire project investment paid back within the first 5 months of pilot deployment.



Zero Friction for Drivers and Field Teams

- **Zero Driver Data Entry:** Drivers focus entirely on road safety; software handles all background analysis automatically.
- **Plugs into Existing Software:** Fits directly into our current dispatch and shop-floor scheduling systems.
- **Instant Shop Alerts:** Maintenance managers receive clear, prioritized notifications without administrative burden.



Automated Dispatch Workflow

Zero manual data entry required from drivers or mechanics

Built-In Operational Guardrails and Fail-Safes



Fail-Safe Backup

Standard inspection schedules remain as an automatic backup if sensor connectivity drops.



Human-in-the-Loop

Master mechanics review and approve all automated service tickets before parts are ordered.



Continuous Quality Check

System accuracy is reviewed monthly against real repair outcomes to stay sharp.

Business, Ethical, & Regulatory Risk Mitigations

Driver Privacy (Ethical Risk)

Mitigation: Strict data isolation. Systems evaluate vehicle engine health ONLY, never tracking individual driver behavior or personal data.

Vendor & Fleet Fairness

Mitigation: Audited across all vehicle brands (Kenworth, Freightliner, Volvo) to ensure equal diagnostic accuracy across the fleet.

Regulatory & DOT Compliance

Mitigation: Fully aligned with Regulatory Administration Board Authority and mandatory DOT inspection standards.

Full Auditability

Mitigation: Every alert stores complete diagnostic records for transparent warranty claims and safety board audits.

Phased Implementation Roadmap

PHASE 1 (WEEKS 1-6)

50-Vehicle Regional Pilot

Launch controlled trial in a logistics corridor to validate real-world shop workflow.

PHASE 2 (MONTHS 3-6)

Fleet-Wide Rollout

Expand software across all regional routes and conduct shop mechanic training sessions.

PHASE 3 (MONTHS 6+)

Automated Parts Ordering

Connect predictive alerts directly to inventory management for automated parts restocking.

Recommendation: Approve 6-Week Regional Pilot

Initiate a 6-Week Regional Pilot (50 Vehicles)

Deploy the trained model in shadow mode across 50 commercial fleet trucks in a single high-density operating region.

Why?

Validates model predictions against real-world shop-floor maintenance logs without disrupting existing fleet operations or risking false alarms.

Connect Telemetry Scoring to CMMS Work Orders

Integrate real-time API output directly into the Computerized Maintenance Management System (CMMS) used by mechanics.

Why?

Automatically routes flagged vehicle alerts as priority shop tickets, ensuring high-risk machines are serviced up to 7 days before an operational failure occurs.

Establish a Monthly Model Drift & Governance Review

Form a cross-functional steering committee (Fleet Operations, Data Science, and IT Security) to review model metrics monthly.

Why?

Continuously monitors input data drift, maintains strict driver data privacy, and ensures breakdown detection accuracy stays consistently above target.